

Tendon-Driven 7-Axis Robot Hand: **THING**

Tendon-Driven 7-Axis Robotic Hand with Real-Time Motion Imitation

[한국어 ↗](#)

공세민 · 김기현 · 백지웅 · 신수진 · 윤정민 · 이정빈

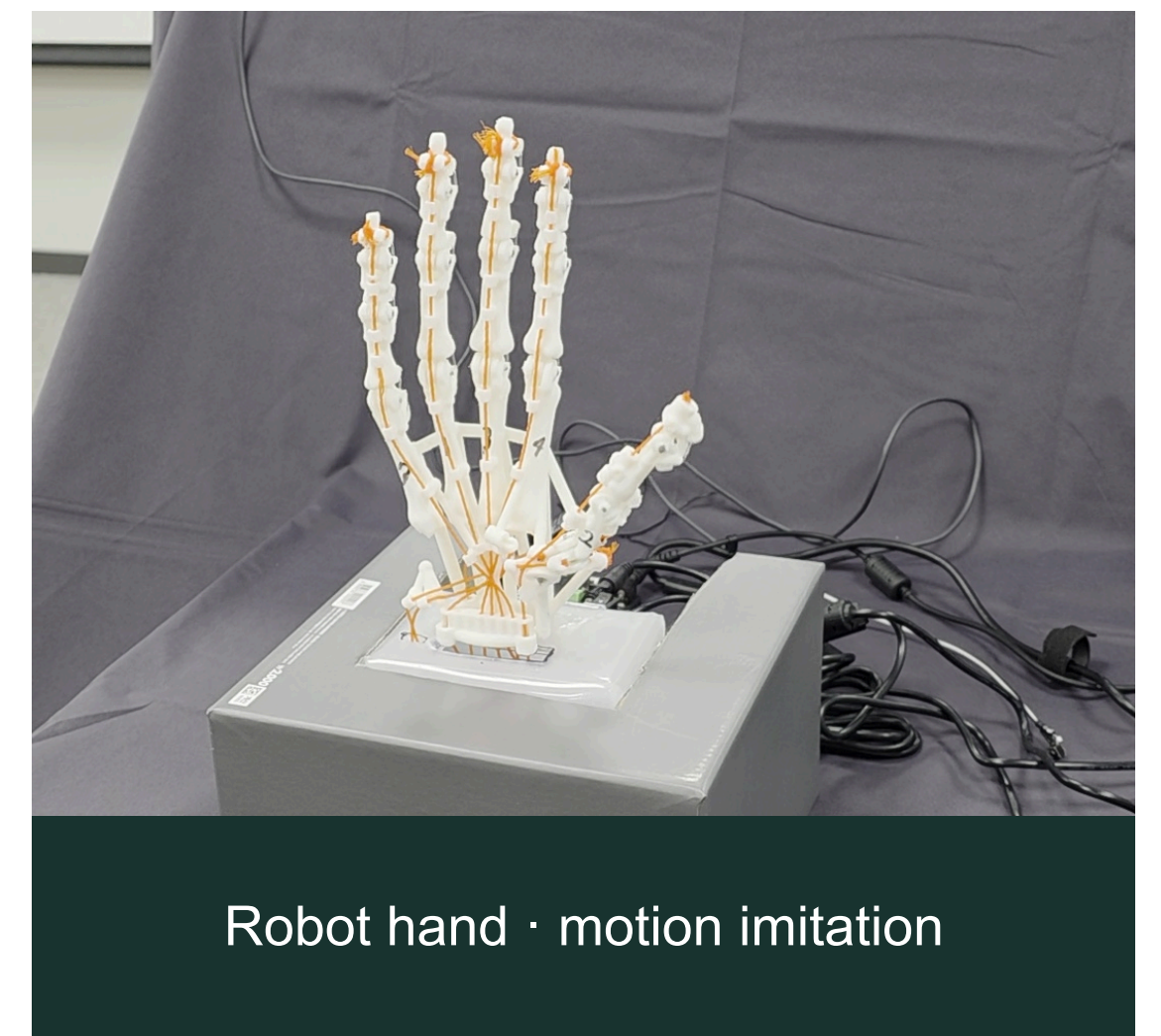
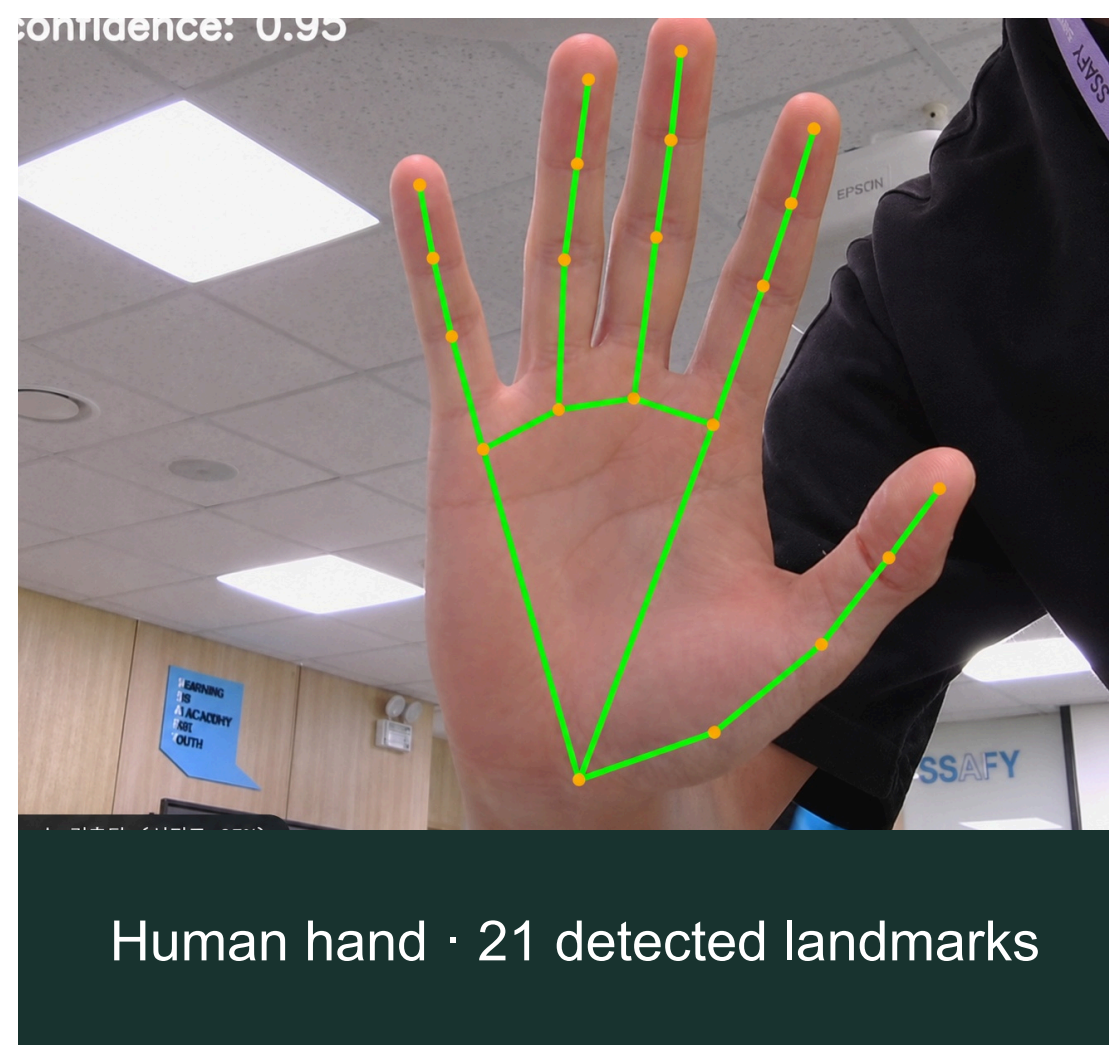
Motion imitation & object grasping

Project background

THING connects human hand-motion recognition, robotic hand actuation and motion recording.

A seven-axis tendon mechanism reproduces motion and grasping while reducing the complexity of driving every joint independently.

The platform supports robot-control education; collected motion data could support future AI training.



From motion to commands

Hand mechanism design

21

Hand landmarks

Detect wrist and finger-joint positions.



7

Normalized control axes

3 thumb axes + 4 finger-flexion axes.



7

Actuation motors

Motors pull tendons to move the fingers.



Coupled finger joints

One tendon couples several joints in an underactuated finger. Joint connections allow flexion while limiting backward motion.

Tendon routing & guides

Guides route the tendon through the finger segments, transmitting pulling force into finger flexion.

Coupled joints prevent motor position from uniquely determining each joint angle. Control uses calibrated motor positions rather than joint-angle feedback.

Finger gestures & object grasping



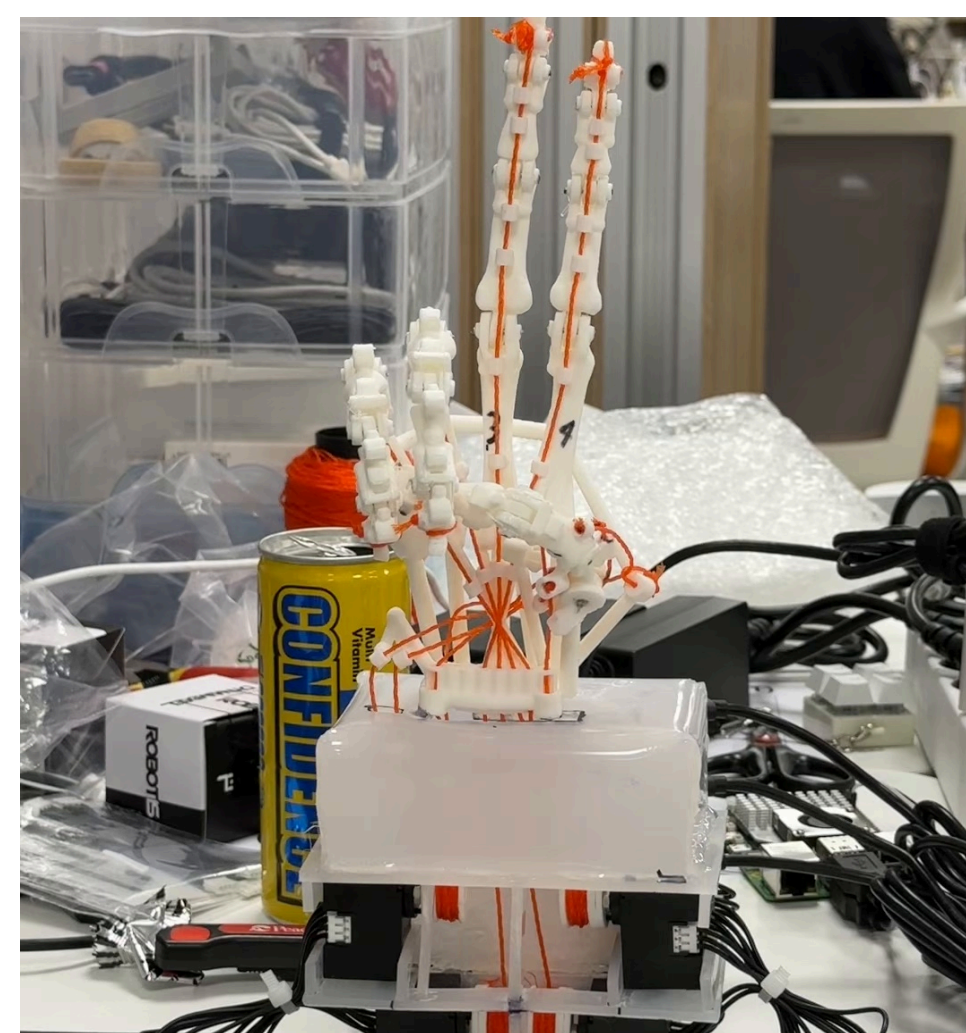
[All demonstrations](#)
[Demo videos ↗](#)



Cylinder · wrapping around a can



Soft object · enveloping grasp



V sign · index & middle extended



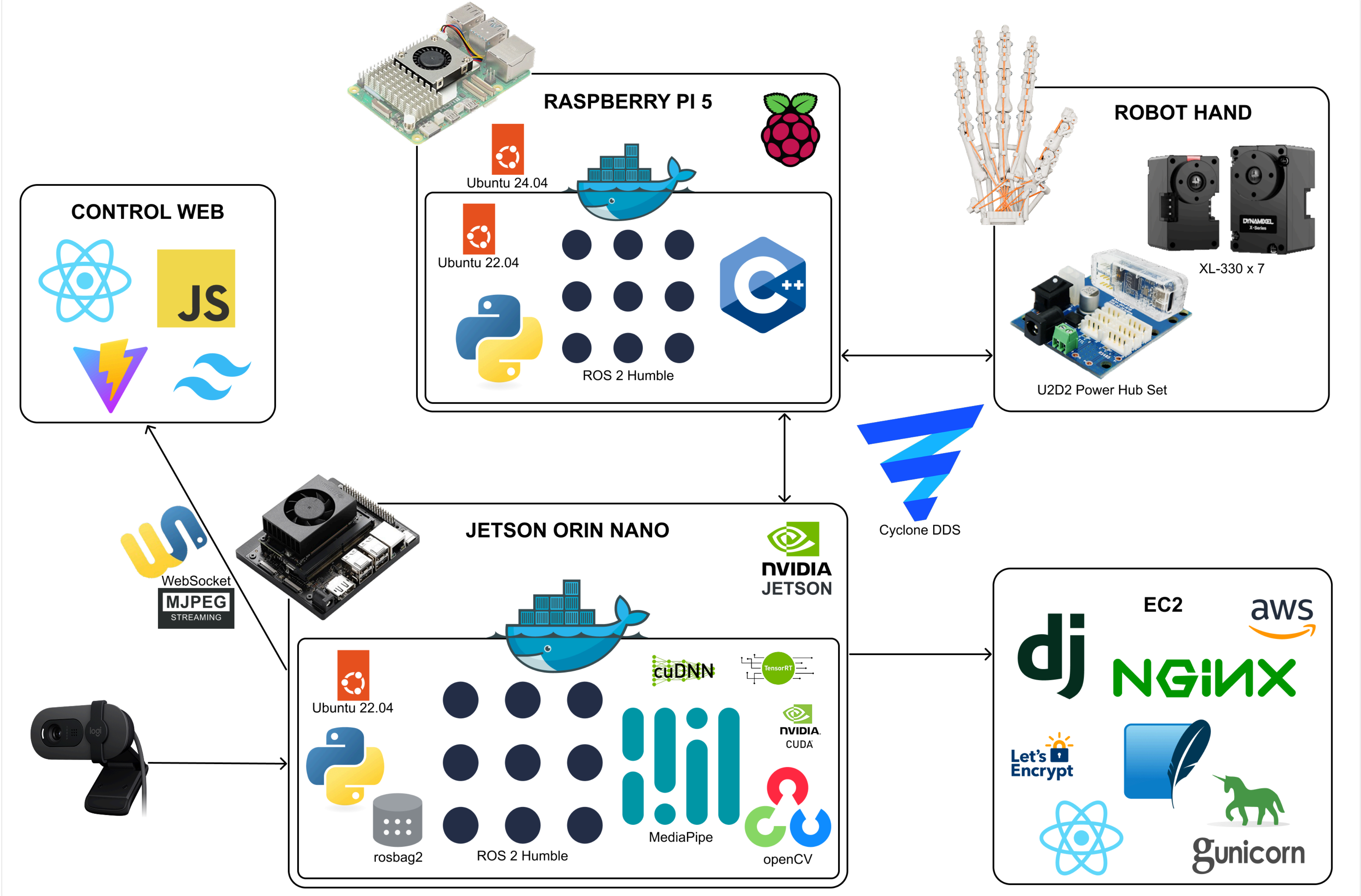
Pinch · thumb & index together

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From camera input to tendon actuation



Jetson handles recognition and recording; Raspberry Pi controls the motors. The control web manages operation and status, with session data retrieved from EC2.

01 Hand image → 21 landmarks

Jetson and MediaPipe extract wrist and finger-joint positions.

`landmarks[21] = {(x, y, z), ...}`

x, y: normalized image coordinates / z: relative depth

02 Landmarks → 7-axis targets

User calibration maps each motion axis to a command from 0 to 1.

Thumb: flexion · opposition · abduction
Index · middle · ring · little: flexion

Deadband → EMA → rate limiting · actuation commands, not joint angles

03 7-axis targets → motor positions

Raspberry Pi validates commands and applies per-motor calibration.

Index: $u \rightarrow 1740 + u \times (4300 - 1740)$

Configuration example · raw ticks / separate thumb-pose mapping

04 Motor rotation → tendon pulling

Seven motors connected via U2D2 pull tendons to move the fingers.

BUS A: 3 motors / BUS B: 3 motors / BUS C: 1 motor

XL330-M288-T × 7 · 3 thumb axes + 4 finger axes